

NORTH SYDNEY GIRLS' HIGH SCHOOL
TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

1997

MATHEMATICS

3U/4U COMMON PAPER

*Time allowed - Two hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES:

- * All questions may be attempted.
- * All questions are of approximately equal value.
- * Part marks for each question are shown in the right hand column.
- * All necessary working must be shown.
- * Marks may be deducted for careless or badly arranged work.
- * **Start each question on a NEW page**
- * This examination is worth 50% of the H.S.C. Assessment Mark
- * Standard integrals are printed on the back page which may be removed for your convenience. Approved calculators may be used.

This is a trial paper ONLY. The content and format of this paper do not necessarily reflect the content and format of the final Higher School Certificate examination paper.

Question 1. (Start a new page)**Marks**

- (a) Evaluate $\int_4^{20} y \, dx$ if $xy = 5$ 2
- (b) Differentiate $y = \tan^{-1}\left(\frac{1}{x}\right)$ 3
- (c) Sketch the curve $y = 2\sin(x + \pi)$ for $0 \leq x \leq 2\pi$ 2
- (d) If $y = ae^{bx}$, show $\frac{d^2y}{dx^2} = b^2y$ where a, b are constants 2
- (e) Solve: $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 6 = 0$ 3

Question 2. (Start a new page)

- (a) State the domain and range of $y = 4\sin^{-1}2x$ and sketch the curve. 3
- (b) Solve: $\cos^2 x - \cos 2x = 0$ for $0 \leq x \leq 2\pi$ 3
- (c) Find the exact value of $\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx$ using the substitution $u = \frac{1}{x}$ 3
- (d) Use $x = 0.5$ to find an approximation to the root of $\cos x = x$ using one application of Newton's method. (Answer correct to two decimal places.) 3

Question 3. (Start a new page)

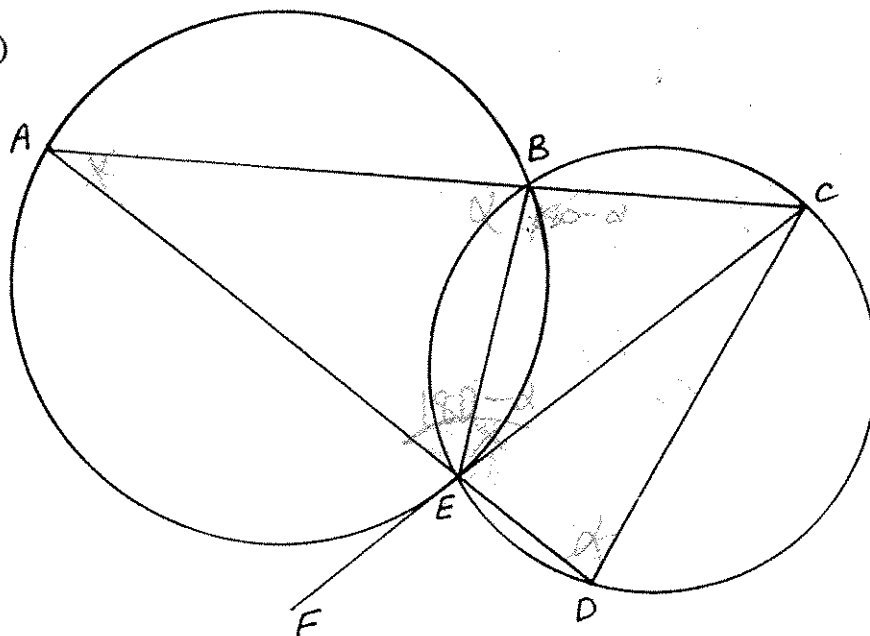
Marks

(a) Solve: $\frac{x^2-3}{2x} > 0$

3

(b)

4



CEF is a tangent to circle AEB

ABC and AED are secants.

- (i) Prove $\triangle ACE$ is similar to $\triangle ECB$
- (ii) Show that $CE = CD$

(c) $P(4p, 2p^2)$ and $Q(4q, 2q^2)$ are two variable points on the parabola $x^2 = 8y$.
R is the point of intersection of the tangents at P and Q.

5

- (i) Show that the co-ordinates of R are $(2[p+q], 2pq)$.
- (ii) Find the cartesian equation of the locus of R, if $p^2 + q^2 = 8$.

Question 4. (Start a new page)

Marks

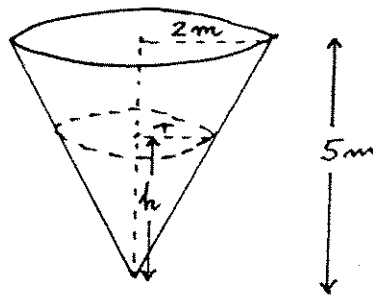
(a) Consider $P(x) = x^4 - x^3 - 3x^2 + 5x - 2$

5

- (i) Show that 1 and -2 are zeros of $P(x)$
- (ii) Using sum and product of roots, or the division algorithm, factorise $P(x)$ into linear factors.

- (b) An inverted right circular cone has height 5m and base radius 2m. Water is flowing from the apex (point) at a constant rate of $0.2m^3/\text{min}$.

5



- (i) If h is the height when the radius is r , show that $r = \frac{2h}{5}$
 - (ii) At height h , show that V , the volume of water is given by
$$V = \frac{4\pi h^3}{75}$$
 - (iii) Hence find the rate at which the water level is falling when the water is 4m deep.
- (c) By letting $t = \tan\left(\frac{\theta}{2}\right)$, prove

2

$$\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan\left(\frac{\theta}{2}\right)$$

Question 5. (Start a new page)**Marks**

- (a) Find the acute angle between the tangents to the curves $y = x^2$ and $y = (x - 2)^2$ at the point of intersection of these two curves. 3
- (b) (i) Show that $T = P + Ae^{kt}$ is a solution of $\frac{dT}{dt} = k(T - P)$ where k , P and A are constants. 6
- (ii) Meat, initially at 14°C is placed in a freezer whose temperature is a constant -10°C . After 25 seconds, the meat is 11°C .
- (α) Show that $A=24$ and $k = -0.005$
- (β) Find (to the nearest minute) when the temperature of the meat will reach -8°C .
- (c) Find the co-ordinates of the point which divides the line joining the points $(-1, 3)$ and $(5, -7)$ externally in the ratio $4:3$ 3

Question 6. (Start a new page)

- (a) (i) Show that $\sqrt{3}\cos x + \sin x$ can be expressed as $2\cos\left(x - \frac{\pi}{6}\right)$ 4
- (ii) Hence state the greatest value of the expression $\sqrt{3}\cos x + \sin x$ and state the smallest positive value of x that gives this maximum value to the expression.
- (b) Consider the graph $y = \frac{x^2}{1-x^2}$ 8
- (i) Write down the domain of this function.
- (ii) Find the turning point and determine its nature.
- (iii) Prove that the function is even.
- (iv) Find $\lim_{x \rightarrow \infty} \frac{x^2}{1-x^2}$
- (v) Sketch the graph.

Question 7. (Start a new page)

Marks

- (a) Evaluate $\int_0^2 \frac{dx}{\sqrt{4-25x^2}}$ 3
- (b) (i) Using the fact $\cos 3x = 4\cos^3 x - 3\cos x$, find the general solutions of the equation $\cos 3x + 2\cos x = 0$ 6
- (ii) What are the smallest and largest solutions for x in part (i) in the interval $0 \leq x \leq 2\pi$?
- (c) Use mathematical induction to prove that $3^{2n+4} - 2^{2n}$ is divisible by 5, for $n \geq 1$. 3

1) (a) $\int_4^{20} y \, dx$ if $xy=5$

$$= \int_4^{20} \frac{5}{x} \, dx$$

$$= 5 [\log x]_4^{20}$$

$$= 5 (\log 20 - \log 4)$$

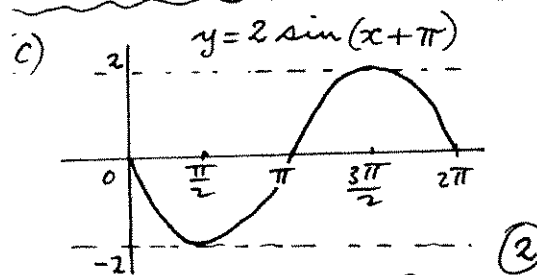
$$= 5 \log 5 \quad (2)$$

2) $y = \tan^{-1}(\frac{1}{x})$

$$\frac{dy}{dx} = \frac{1}{1+(\frac{1}{x})^2} \times -\frac{1}{x^2}$$

$$= \frac{x^2}{x^2+1} \times -\frac{1}{x^2}$$

$$= \frac{-1}{x^2+1} \quad (3)$$



4) (d) $y = ae^{bx}$

$$\frac{dy}{dx} = abe^{bx}$$

$$\frac{d^2y}{dx^2} = ab^2e^{bx}$$

$$= b^2y \quad (2)$$

5) (e) $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 6 = 0$

Set $m = x^{\frac{1}{3}}$

$$\therefore m^2 + m - 6 = 0$$

$$(m+3)(m-2) = 0$$

$$\therefore m = -3, m = 2$$

$$\therefore x^{\frac{1}{3}} = -3 \quad \therefore x^{\frac{1}{3}} = 2$$

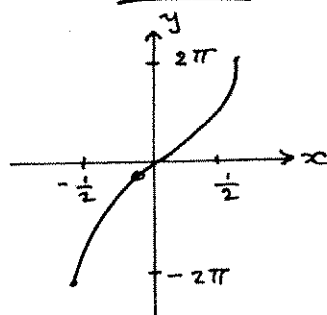
$$\therefore x = -27 \quad \therefore x = 8 \quad (3)$$

2) (a) D: $-1 \leq 2x \leq 1$

$$\therefore -\frac{1}{2} \leq x \leq \frac{1}{2}$$

R: $4x(-\frac{\pi}{2}) \leq y \leq 4x(\frac{\pi}{2})$

$$-2\pi \leq y \leq 2\pi$$



6) (b) $\cos^2 x - \cos 2x = 0$

$$\therefore \cos^2 x - (2\cos^2 x - 1) = 0$$

$$\therefore \cos^2 x - 2\cos^2 x + 1 = 0$$

$$\therefore \cos^2 x = 1$$

$$\therefore \cos x = \pm 1$$

$$\therefore x = 0, \pi, 2\pi \quad (3)$$

7) (c) $I = \int_1^2 \frac{e^{\frac{1}{x}}}{x^2} \, dx$

Let $u = \frac{1}{x} \quad \begin{cases} x=1, u=1 \\ x=2, u=\frac{1}{2} \end{cases}$

$$\therefore \frac{du}{dx} = -\frac{1}{x^2}$$

$$\therefore -du = \frac{dx}{x^2}$$

$$\therefore I = \int_1^{\frac{1}{2}} e^u \, du$$

$$\therefore I = -[e^u]^{\frac{1}{2}}_1$$

$$= -\{e^{\frac{1}{2}} - e^1\}$$

$$= e - e^{\frac{1}{2}} \quad (3)$$

8) (d) $a_1 = a - \frac{f(a)}{f'(a)}$

$$a = 0.5$$

$$f(a) = \cos(0.5) - (0.5)$$

$$f'(a) = -\sin(0.5) - 1$$

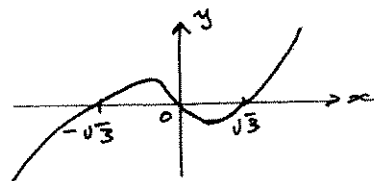
$$a_1 = 0.76 \quad (3)$$

3) (a) $\frac{x^2-3}{2x} > 0$

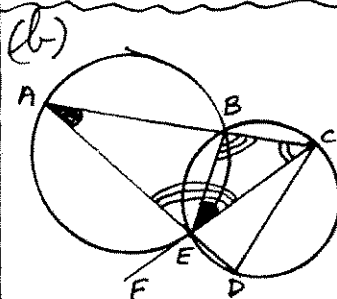
$$\therefore \frac{(x^2-3)x^2}{2x} > 0$$

$$\therefore x(x^2-3) > 0$$

$$x(x+\sqrt{3})(x-\sqrt{3}) > 0$$



$$\therefore -\sqrt{3} < x < 0, x > \sqrt{3} \quad (3)$$



(i) $\widehat{BAE} = \widehat{BEC}$ (L in alt. seg.)

$\widehat{BCE}$ is common

$\therefore \widehat{AEC} = \widehat{EBC}$ (L sum of Δ 's)

$\therefore \Delta ACE \parallel \Delta ECB$ (equiang)

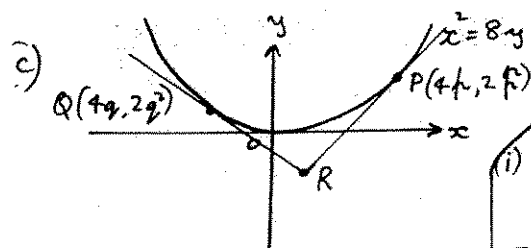
(ii) $\widehat{EDC} = 180 - \widehat{EBC}$ (opp L, cyclic quad)

$\widehat{CED} = 180 - \widehat{AEC}$ (str. L)

$\therefore \widehat{EDC} = \widehat{CED}$ ($\widehat{AEC} = \widehat{EBC}$ from (i))

$\therefore \Delta CED$ is isosceles

$\therefore CE = CD$ (opp L's in iso Δ) (4)



i) $x^2 = 8y$

$$y = \frac{x^2}{8}$$

$$\frac{dy}{dx} = \frac{x}{4}$$

$$= 4t \frac{dy}{dx} = \frac{4t}{4} = t$$

ing at P is:

$$y - 2t^2 = t(x - 4t)$$

$$y - 2t^2 = tx - 4t^2$$

$$y = tx - 2t^2 \dots (1)$$

Similarly tang at Q is:

$$y = qx - 2q^2 \dots (2)$$

Solving (1) and (2)

$$tx - 2t^2 = qx - 2q^2$$

$$x(t - q) = 2(t^2 - q^2)$$

$$\therefore x = 2(t + q)$$

Sub in (1)

$$y = t \cdot 2(t + q) - 2t^2$$

$$y = 2t^2 + 2tq - 2t^2$$

$$y = 2tq$$

R is $(2[t+q], 2tq)$.

Let R be (x, y)

$$x = 2(t+q) \text{ and } y = 2tq$$

$$\text{or } (t+q)^2 = t^2 + q^2 + 2tq$$

$$\left(\frac{x}{2}\right)^2 = 8 + y$$

$$\therefore \frac{x^2}{4} = 8 + y$$

$$\therefore x^2 = 4(y + 8) \quad (5)$$

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4(a)

$$P(x) = x^4 - x^3 - 3x^2 + 5x - 2$$

$$P(1) = 1 - 1 - 3 + 5 - 2 = 0$$

$$P(-2) = 16 + 8 - 12 - 10 - 2 = 0$$

(ii) Let roots be $\alpha, \beta, 1, -2$

$$\therefore \alpha + \beta + 1 - 2 = 1$$

$$\therefore \alpha + \beta = 2 \dots (1)$$

$$\text{and } (\alpha)(\beta)(1)(-2) = -2$$

$$\therefore -2\alpha\beta = -2$$

$$\therefore \alpha\beta = 1 \dots (2)$$

Solving (1) and (2)

$$\alpha(2 - \alpha) = 1$$

$$2\alpha - \alpha^2 = 1$$

$$\alpha^2 - 2\alpha + 1 = 0$$

$$(\alpha - 1)^2 = 0$$

$$\therefore \alpha = 1$$

$$\beta = 1$$

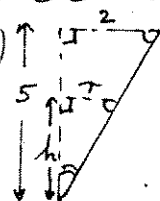
$$\therefore P(x) = (x-1)^3(x+2)$$

OR

$$\begin{array}{r} x^2 - 2x + 1 \\ x^2 + x - 2 \overline{) x^4 - x^3 - 3x^2 + 5x - 2} \\ \underline{x^4 + x^3 - 2x^2} \\ -2x^3 - x^2 + 5x \\ \underline{-2x^3 - 2x^2 + 4x} \\ x^2 + x - 2 \\ \underline{x^2 + x - 2} \\ 0 \end{array}$$

$$\therefore P(x) = (x-1)^3(x+2) \quad (5)$$

(b)(i) By similar Δ



$$\frac{r}{2} = \frac{h}{5}$$

$$\therefore r = \frac{2h}{5}$$

$$(ii) V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi \left(\frac{2h}{5}\right)^2 h$$

$$= \frac{4\pi h^3}{75}$$

$$(iii) \frac{dh}{dt} = ? \quad V = \frac{4\pi h^3}{75}$$

$$\frac{dV}{dt} = 0.2 \quad \therefore \frac{dV}{dh} = \frac{4\pi h^2}{25}$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$= \frac{25}{4\pi h^2} \times 0.2$$

$$= \frac{5}{4\pi h^2}$$

(5)

$$\text{when } h = 4: \frac{dh}{dt} = \frac{5}{4\pi \cdot 16} = \frac{5}{64\pi} \text{ m/min}$$

$$(c) \text{ Prove (let } t = \tan(\frac{\theta}{2}))$$

$$\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan\left(\frac{\theta}{2}\right)$$

$$\text{L.H.S.} = \frac{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}$$

$$= \frac{1+t^2+2t-1+t^2}{1+t^2+2t+1-t^2}$$

$$= \frac{2t+2t^2}{2+2t}$$

$$= \frac{2t(1+t)}{2(1+t)}$$

$$= t$$

$$= \text{R.H.S.}$$

(2)

12

$$5(a) y = x^2$$

$$y = (x-2)^2$$

$$\therefore x^2 = x^2 - 4x + 4$$

$$4x = 4$$

$$\underline{x = 1}, \underline{y = 1}$$

Pt of int. is (1, 1)

For $y = x^2$

$$\frac{dy}{dx} = 2x$$

$$\text{At } x = 1: \frac{dy}{dx} = 2$$

$$\therefore m_1 = 2$$

For $y = (x-2)^2$

$$\frac{dy}{dx} = 2(x-2)$$

at $x=1$ $\frac{dy}{dx} = -2$

$$\therefore m_2 = -2$$

our $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$$= \left| \frac{2 + 2}{1 - 4} \right|$$

$$= \left| \frac{4}{-3} \right|$$

$$= \frac{4}{3}$$

$$\therefore \theta = 53^\circ 8' \quad (3)$$

(i) $T = P + Ae^{kt}$

$$\frac{dT}{dt} = kAe^{kt}$$

but $Ae^{kt} = T - P$

$$\therefore \frac{dT}{dt} = k(T - P)$$

i) (x) $T = P + Ae^{kt}$

$$\left. \begin{array}{l} t=0 \\ T=-10 \\ T=14 \end{array} \right\} \begin{array}{l} 14 = -10 + A \\ A = 24 \end{array}$$

$$\left. \begin{array}{l} t=25 \\ T=11 \end{array} \right\} \begin{array}{l} 11 = -10 + 24e^{25k} \\ \frac{21}{24} = e^{25k} \end{array}$$

$$\therefore k = \frac{1}{25} \log\left(\frac{21}{24}\right)$$

$$\div -0.005$$

$$\left. \begin{array}{l} T=-8 \\ t=? \end{array} \right\} \begin{array}{l} -8 = -10 + 24e^{-0.005t} \\ \frac{2}{24} = e^{-0.005t} \end{array}$$

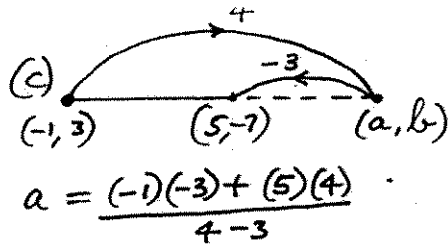
$$\frac{2}{24} = e^{-0.005t}$$

$$\therefore \log\left(\frac{1}{12}\right) = -0.005t$$

$$\therefore t = \frac{\log\left(\frac{1}{12}\right)}{-0.005}$$

$$= 496.98 \text{ sec}$$

$$\div 8 \text{ min} \quad (6)$$



$$a = \frac{(-1)(-3) + (5)(4)}{4 - 3}$$

$$= 23$$

$$b = \frac{(3)(-3) + (-7)(4)}{4 - 3}$$

$$= -37$$

Pt is $(23, -37)$ (3)

6) (a) (i) Let $\sqrt{3}\cos x + \sin x$

$$\equiv A\cos\alpha\cos x + A\sin\alpha\sin x$$

$$\therefore A\cos\alpha = \sqrt{3}$$

$$A\sin\alpha = 1$$

$$\therefore \tan\alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6}$$

and $A^2(\cos^2\alpha + \sin^2\alpha) = 4$

$$\therefore A = 2$$

$$\therefore \sqrt{3}\cos x + \sin x = 2\cos\left(x - \frac{\pi}{6}\right)$$

(ii) greatest value is 2

x value is $\frac{\pi}{6}$ (4)

(b) $y = \frac{x^2}{1-x^2}$

(i) D: all real x, $x \neq \pm 1$

(ii) $y = \frac{x^2}{1-x^2}$

$$\frac{dy}{dx} = \frac{(1-x^2)2x - x^2(-2x)}{(1-x^2)^2}$$

$$= \frac{2x - 2x^3 + 2x^3}{(1-x^2)^2}$$

$$= \frac{2x}{(1-x^2)^2}$$

$$= 0 \text{ when } x = 0$$

$\therefore$ stat. pt at $(0, 0)$

$f'(x) < 0, f'(0) = 0, f'(1) > 0$

$\therefore$ Min. + p. at $(0, 0)$

(iii) $f(-x) = \frac{(-x)^2}{1-(-x)^2}$

$$= \frac{x^2}{1-x^2}$$

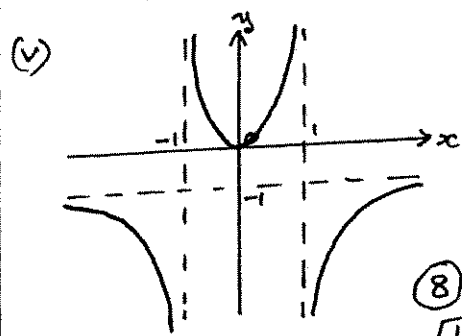
$$= f(x)$$

(iv) $\lim_{x \rightarrow \infty} \frac{x^2}{1-x^2}$

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2}}{\frac{1}{x^2} - \frac{x^2}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{x^2} - 1}$$

$$= -1 \quad \left(\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0 \right)$$



7) (a) $\int_0^{\frac{2}{5}} \frac{dx}{\sqrt{4-25x^2}}$

$$= \frac{1}{5} \int_0^{\frac{2}{5}} \frac{dx}{\sqrt{\left(\frac{2}{5}\right)^2 - x^2}}$$

$$= \frac{1}{5} \left[\sin^{-1}\left(\frac{x}{\frac{2}{5}}\right) \right]_0^{\frac{2}{5}}$$

$$= \frac{1}{5} \left[\sin^{-1} \frac{5x}{2} \right]_0^{\frac{2}{5}}$$

$$= \frac{1}{5} \left\{ \sin^{-1} 1 - \sin^{-1} 0 \right\}$$

$$= \frac{1}{5} \left(\frac{\pi}{2} - 0 \right)$$

$$= \frac{\pi}{10}$$

(3)

(b) (i) If $\cos 3x = 4\cos^3 x - 3\cos x$
then $\cos 3x + 2\cos x = 0$ becomes

$$4\cos^3 x - 3\cos x + 2\cos x = 0$$

$$\therefore 4\cos^3 x - \cos x = 0$$

$$\cos x (4\cos^2 x - 1) = 0$$

$$\cos x (2\cos x - 1)(2\cos x + 1) = 0$$

$$\therefore \cos x = 0, \pm \frac{1}{2}$$

General solutions:

$$\cos x = 0:$$

$$x = 2n\pi \pm \cos^{-1} 0$$

$$x = 2n\pi \pm \frac{\pi}{2}$$

$$\cos x = \frac{1}{2}:$$

$$x = 2n\pi \pm \cos^{-1}(\frac{1}{2})$$

$$x = 2n\pi \pm \frac{\pi}{3}$$

$$\cos x = -\frac{1}{2}:$$

$$x = 2n\pi \pm \cos^{-1}(-\frac{1}{2})$$

$$= 2n\pi \pm (\pi - \cos^{-1}(\frac{1}{2}))$$

$$= 2n\pi \pm (\pi - \frac{\pi}{3})$$

$$x = 2n\pi \pm \frac{2\pi}{3}$$

(ii) smallest:

$$(n=0 \text{ in } x = 2n\pi + \frac{\pi}{3})$$

$$x = \frac{\pi}{3}$$

largest:

$$(n=1 \text{ in } x = 2n\pi - \frac{\pi}{3})$$

$$x = \frac{5\pi}{3}$$

(6)

$$(c) 3^{2n+4} - 2^{2n}$$

$$n=1: 3^6 - 2^2$$

$$= 725$$

$\therefore$ divisible by 5

True for $n=1$

Assume true for $n=k$.

$$\therefore 3^{2k+4} - 2^{2k} = 5M$$

To prove true for $n=k+1$

$$\text{Now } 3^{2(k+1)+4} - 2^{2(k+1)}$$

$$= 3^{2k+6} - 2^{2k+2}$$

$$= 3^{(2k+4)+2} - 2^{2k+2}$$

$$= 9 \cdot 3^{2k+4} - 4 \cdot 2^{2k}$$

$$= 9(5M + 2^{2k}) - 4 \cdot 2^{2k}$$

$$= 45M + 9 \cdot 2^{2k} - 4 \cdot 2^{2k}$$

$$= 45M + 5 \cdot 2^{2k}$$

$$= 5[9M + 2^{2k}]$$

Having assumed true for $n=k$,

proven true for $n=k+1$

BUT true for $n=1$

$\therefore$ true for $n=2$

$\therefore$ True for $n=3, 4, 5, \dots, \infty$.

(3)

12

$$\text{Total} = \underline{84}$$